VTPEH6270_CP04

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Introduction

This report explores the association between physical activity and the prevalence of any poor mental health days using the NYSDOH BRFSS 2023 dataset. The analysis focuses on whether adults who reported any exercise are less likely to report poor mental health days compared with those who reported no exercise.

Objective and Approach

Research Question

Do adults who reported any physical activity or exercise have a lower prevalence of any poor mental health days than those who reported no exercise?

Data Description

```
brfss <- read.csv("NYSDOH_BRFSS_SurveyData_2023_.csv",
                  stringsAsFactors = FALSE)

dim(brfss)
```

```
## [1] 17349   370
```

```
names(brfss)[1:20]
```

```
##  [1] "X_STATE"  "IDATE"    "IMONTH"   "IDAY"     "SEQNO"    "X_PSU"
##  [7] "CELPHON1" "NUMADULT" "LANDSEX2" "LNDSXBRT" "CELLSEX2" "CELSXBRT"
## [13] "HHADULT"  "SEXVAR"   "GENHLTH"  "PHYSHLTH" "MENTHLTH" "POORHLTH"
## [19] "PRIMINS1" "PERSDOC3"
```

```
names(brfss)[grep("EXER|PHYACT|ACTIV|EXERC", names(brfss), ignore.case = TRUE)]
```

```
## [1] "EXERANY2" "EXEROFT1" "EXERHMM1" "EXEROFT2" "EXERHMM2" "ARTHEXER"
```

```
dat <- brfss %>%
  select(MENTHLTH, EXERANY2)

head(dat)
```

```
##         MENTHLTH EXERANY2
## 1           None      Yes
## 2 Number of days      Yes
## 3 Number of days      Yes
## 4           None      Yes
## 5 Number of days      Yes
## 6           None      Yes
```

```
table(dat$MENTHLTH)
```

```
##
## Don\x92t know/Not sure                   None         Number of days
##                    341                   9905                   6995
##                Refused
##                    108
```

```r
dat2 <- dat %>%
  filter(MENTHLTH %in% c("None", "Number of days"),
         EXERANY2 %in% c("Yes", "No")) %>%
  mutate(
    poor_mental = ifelse(MENTHLTH == "Number of days", 1, 0),
    exercise = ifelse(EXERANY2 == "Yes", 1, 0)
  )

head(dat2)
```

```
##         MENTHLTH EXERANY2 poor_mental exercise
## 1           None      Yes           0        1
## 2 Number of days      Yes           1        1
## 3 Number of days      Yes           1        1
## 4           None      Yes           0        1
## 5 Number of days      Yes           1        1
## 6           None      Yes           0        1
```

```r
table(dat2$poor_mental)
```

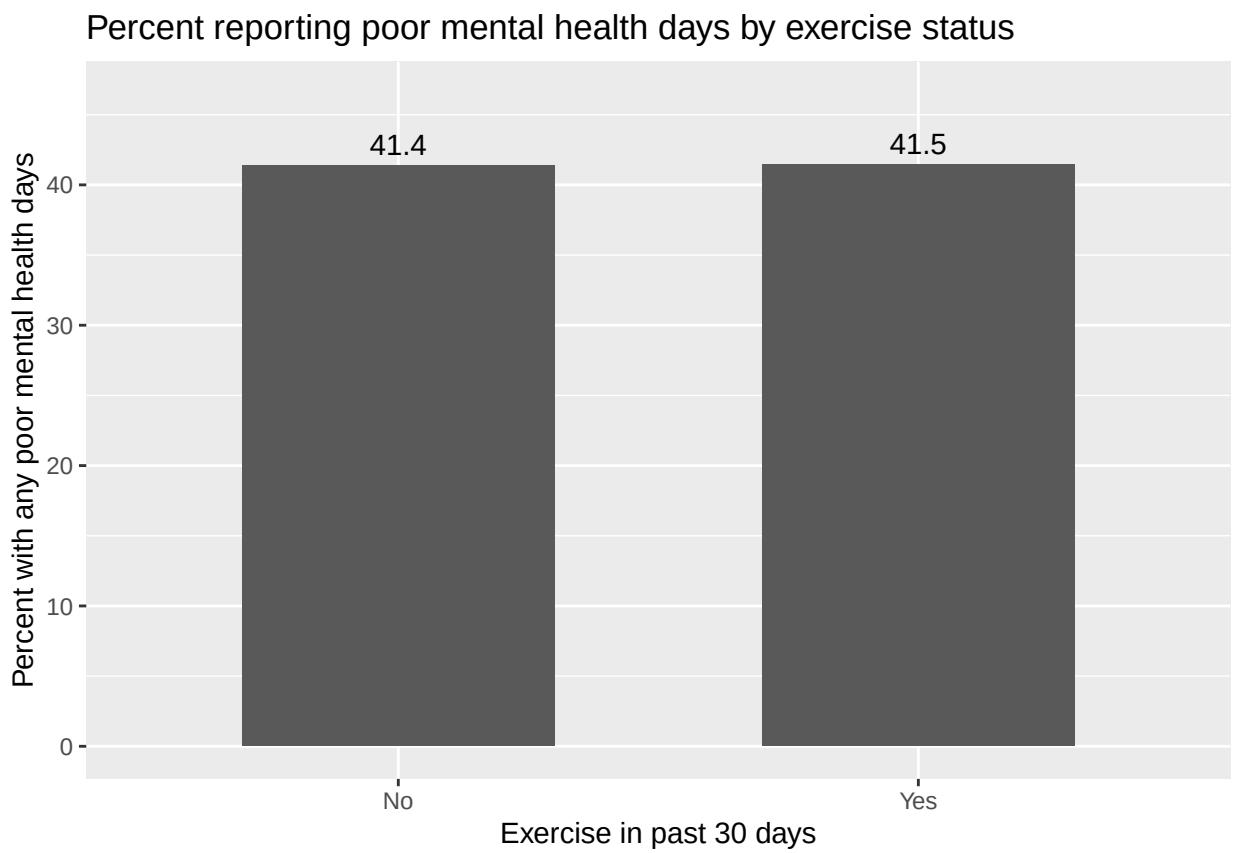
```
##
##    0    1
## 9875 6987
```

```r
table(dat2$exercise)
```

```
##
##     0     1
##  4054 12808
```

Data Visualization

```r
plot_dat <- dat2 %>%
  group_by(EXERANY2) %>%
  summarise(prop_poor_mental = mean(poor_mental)) %>%
  mutate(percent = prop_poor_mental * 100)

ggplot(plot_dat, aes(x = EXERANY2, y = percent)) +
  geom_col(width = 0.6) +
  geom_text(aes(label = round(percent, 1)), vjust = -0.5) +
  labs(
    x = "Exercise in past 30 days",
    y = "Percent with any poor mental health days",
    title = "Percent reporting poor mental health days by exercise status"
  ) +
  ylim(0, max(plot_dat$percent) + 5)
```



The bar chart shows the percentage of respondents who reported any poor mental health days in the past 30 days by exercise status. The percentage appears very similar between individuals who reported exercise and those who did not. This suggests that the prevalence of poor mental health days does not differ substantially between the two groups in this dataset.

Description of Plausible Relationships

A plausible relationship is that individuals who engage in physical activity may have a lower probability of reporting poor mental health days. Since both variables are binary indicators, the relationship can be described as a difference in probabilities between the exercise group and the no-exercise group.

Parameters of Interest

The main parameters of interest are the probability of reporting poor mental health days among respondents who exercised and the probability among those who did not exercise. The key effect size is the difference between these two probabilities.

```
param_table <- tibble(
  Parameter = c("p0", "p1", "effect size"),
  Meaning = c(
    "Probability of poor mental health days in the no-exercise group",
    "Probability of poor mental health days in the exercise group",
    "Difference between the two probabilities"
  )
)
```

```r
kable(param_table)
```

Parameter	Meaning
p0	Probability of poor mental health days in the no-exercise group
p1	Probability of poor mental health days in the exercise group
effect size	Difference between the two probabilities

Simulation

Simulation Basis

The simulation assumes that the probability of reporting poor mental health days may differ between individuals who exercise and those who do not exercise. The outcome is treated as a binary variable indicating whether a respondent reports any poor mental health days. The simulation generates synthetic datasets using specified probabilities for the exercise and no-exercise groups.

Simulation Output

Each simulation generates a dataset with two variables: exercise status and poor mental health outcome. The primary output of interest is the estimated difference in proportions of poor mental health days between the exercise and no-exercise groups.

Simulation Function

```r
simulate_data <- function(n, p0, p1) {

  exercise <- rbinom(n, 1, 0.5)

  poor_mental <- ifelse(
    exercise == 1,
    rbinom(n, 1, p1),
    rbinom(n, 1, p0)
  )

  data.frame(
    exercise = exercise,
    poor_mental = poor_mental
  )
}
```

```r
test_sim <- simulate_data(
  n = 1000,
  p0 = 0.45,
  p1 = 0.40
)

head(test_sim)
```

```
##   exercise poor_mental
## 1        0           1
## 2        1           1
## 3        0           1
## 4        1           1
## 5        0           0
## 6        1           0
```

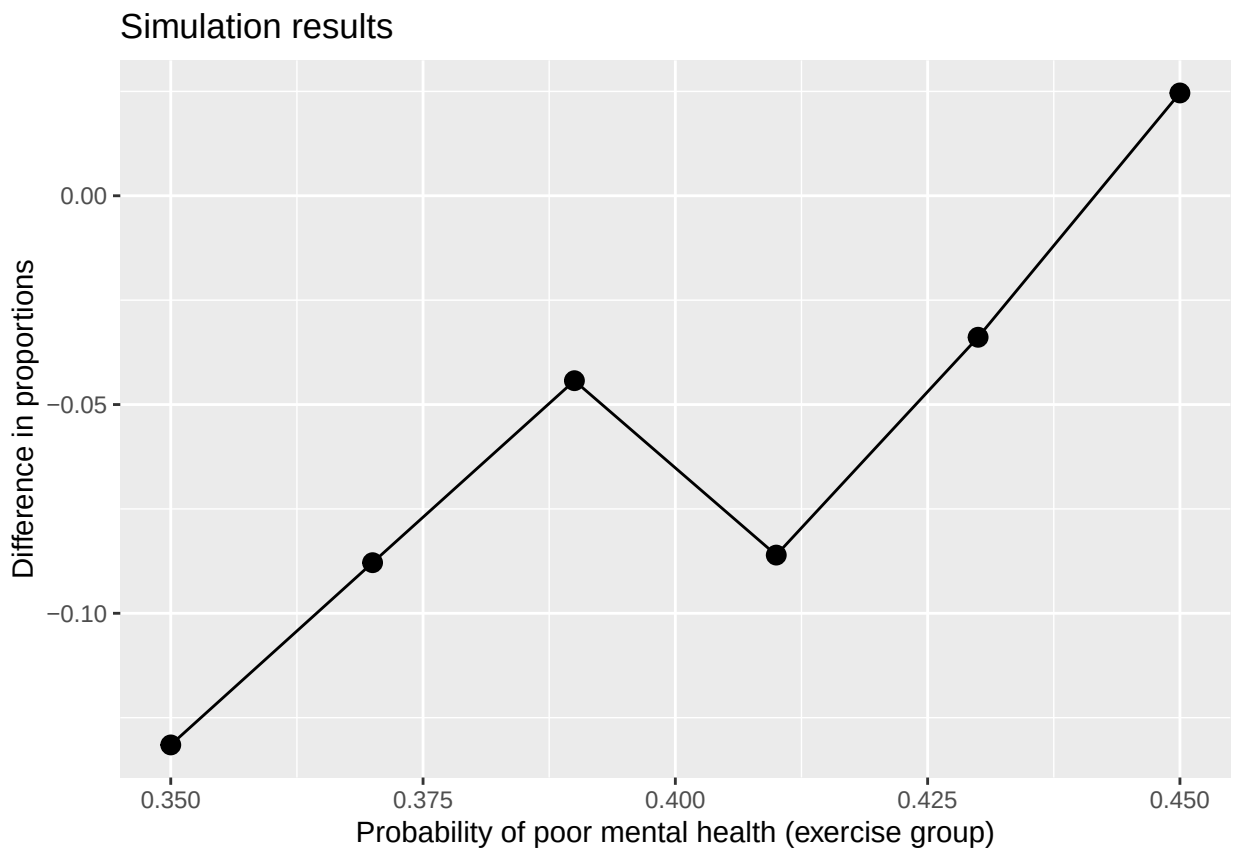
Simulation Automation

```r
set.seed(123)

sim_results <- map_dfr(
  seq(0.35, 0.45, by = 0.02),
  function(p1) {

    sim <- simulate_data(
      n = 1000,
      p0 = 0.45,
      p1 = p1
    )

    diff <- sim %>%
      group_by(exercise) %>%
      summarise(p = mean(poor_mental)) %>%
      summarise(diff = diff(p)) %>%
      pull(diff)

    tibble(
      p1 = p1,
      difference = diff
    )

  }
)

sim_results
```

```
## # A tibble: 6 x 2
##      p1 difference
##   <dbl>      <dbl>
## 1  0.35    -0.132
## 2  0.37    -0.0879
## 3  0.39    -0.0443
## 4  0.41    -0.0861
## 5  0.43    -0.0339
## 6  0.45     0.0246
```

Visualization

```r
ggplot(sim_results, aes(x = p1, y = difference)) +
  geom_point(size = 3) +
  geom_line() +
  labs(
    x = "Probability of poor mental health (exercise group)",
    y = "Difference in proportions",
    title = "Simulation results"
  )
```



Interpretation

The simulation illustrates how changes in the probability of poor mental health days among individuals who exercise affect the difference between the exercise and no-exercise groups. When the probability of poor mental health is lower for the exercise group, the difference in proportions becomes more negative, indicating better mental health outcomes among those who exercise. This demonstrates how the assumed effect size influences the observed association between physical activity and poor mental health days.

AI Use Disclosure Statement

I used ChatGPT to help with understanding some R coding steps and troubleshooting errors during the analysis. I reviewed and edited all code, figures, and written explanations to ensure they accurately reflect my own understanding of the assignment.